SpeechSpectrum: A Linguistic Fidelity Spectrum for Accountable Speech-to-Text

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Extended Abstract

Speech-to-text (STT) systems are deeply embedded in everyday technologies – from meeting transcription and medical documentation to legal proceedings and accessibility tools – yet the standard assumption is to treat transcription as a technical accuracy problem with a single "correct" output. This assumption obscures a consequential social reality: speech can be legitimately represented in multiple ways, and different users, tasks, and communities require different balances of linguistic fidelity, concision, and emphasis. A court reporter documenting testimony needs to preserve every hesitation and false start; a physician reviewing triage notes needs concise bullet points; a Deaf user relies on rich paralinguistic annotations to access conversational nuance [1]. These are not edge cases – they represent the diversity of contexts in which STT systems are deployed at scale. When systems impose representational choices without user input, they encode particular linguistic norms and communicative values, suppress meaningful variation, and deny users – especially those from marginalized communities – agency over how their speech is rendered. We first ask: *How do users want speech to be represented, and is it task-dependent?* And in a time when LLMs are increasingly used for social simulation [4], we ask: *Are LLMs capable of making decisions about how users want speech to be represented?*

In service of these questions, we introduce **SpeechSpectrum, a framework that reconceptualizes STT conversion as *cross-modal translation* along a continuous spectrum of representational fidelity (Figure 1).** Drawing on translation studies and linguistics of orality [3, 5], we define *verbatimicity* – the degree to which textual output preserves structural, lexical, and paralinguistic characteristics of speech – as a designable parameter rather than an optimization target. The spectrum ranges from paralinguistic transcript through verbatim, non-verbatim, standardized, and enhanced representations, down to bullet points and keyword extraction. Each level serves distinct communicative functions and user needs that cannot be fully substituted by other positions on the spectrum; the appropriate choice is inherently contextual and normative.

Three principles organize the framework shown in Figure 1:

(1) *User-controlled Fidelity*, where users specify their preferred representation level rather than accepting system defaults;

(2) *Context-Dependent Optimization*, where optimal fidelity varies by domain, task, and user goals rather than abstract accuracy norms; and

(3) *Cross-Modal Translation*, which treats fidelity as a deliberate representational choice rather than minimizing transcription error against a single correct target.

To support this framework, **we developed an open-source Python package** (`speechspectrum`) that translates speech along this continuum. We validate SpeechSpectrum through two complementary empirical studies:

(1) Human-Subjects Study Evaluating Contextual Fidelity Preferences. In the first study (N = 52) with participants from legal, medical, and business professional backgrounds, we built an interactive prototype instantiating four fidelity levels – Verbatim, Non-Verbatim, Enhanced, and Bullet Points – and asked participants to select the representation that best supported answering six domain-specific task questions. Results demonstrate that fidelity preferences are strongly context- and task-dependent rather than universal (Figure 2a). For example, verbatim transcripts were preferred for legal confidence assessment, while bullet points dominated for identifying medical symptoms and business action items. Chi-squared goodness-of-fit analyses confirm statistically significant departures from uniform preference distributions across five of six questions, with Cramér's V ranging from 0.29 to 0.70 (Table 1). This indicates a moderate to strong relationship between the specific task being performed and the chosen level of transcription fidelity. **These results directly challenge the assumption that a single transcription standard can serve all users and contexts.**

(2) An LLM Audit Testing Whether Models Can Reliably Proxy These Human Choices. In our second study, we examine whether large language models (LLMs) can serve as scalable proxies for human preference in STT evaluation – a question with broad implications for computational social science methodology [2, 4, 6]. We constructed N=52 LLM personas aligned with participant demographics (processional background, STEM experience, domain expertise) and administered the same six questions using GPT-5.1, running three rounds at high temperatures ($\tau = \{0.5, 1.0, 1.5\}$) to allow for variance and properly assess response stability. Not only did LLMs match human top-choice preferences in only 3 of 6 tasks, but also **LLMs consistently produced far more concentrated and homogenized preference distribution** – frequently converging exclusively on a single representation type regardless of task nuance (Figure 2b) – **whereas human preferences remained contextually diverse, pluralistic, and distributed across multiple fidelity levels**. LLMs amplified professional role stereotypes to extremes rather than capturing the flexible, context-sensitive reasoning that characterized human participants. This finding underscores that human-centered studies remain strictly necessary for setting verbatimicity parameters in STT systems. Furthermore, it has implications well beyond the speech technology: it cautions researchers against using LLMs as proxies for heterogeneous human preferences in evaluation frameworks for sociotechnical systems more broadly, and highlights the continued necessity of grounding preference modeling in actual human data.

We derive nine design recommendations for user-centered STT systems, organized around four themes: supporting multi-fidelity interactions so users can navigate across verbatimicities rather than receiving a single output; incorporating task-aware (rather than coarse domain-level) defaults where plurality preferences are observable; providing educational scaffolding to help users reason about fidelity trade-offs for tasks with more diffuse preferences; and developing alignment-based evaluation metrics that assess fidelity relative to specific user goals rather than a single static baseline.

More broadly, **SpeechSpectrum repositions fidelity control as a core site of algorithmic accountability**. By open-sourcing our `speechspectrum` package, we provide system designers with an assistive tool to break away from rigid defaults, while empowering users to dictate their own positions on the verbatimicity spectrum. This redistribution of agency has direct implications for whose speech patterns are accommodated, normalized away, or treated as legitimate. By making representational choices explicit and user-controllable, SpeechSpectrum provides both a conceptual reorientation for the field and a practical mechanism for ensuring diverse communicative needs are met.

References

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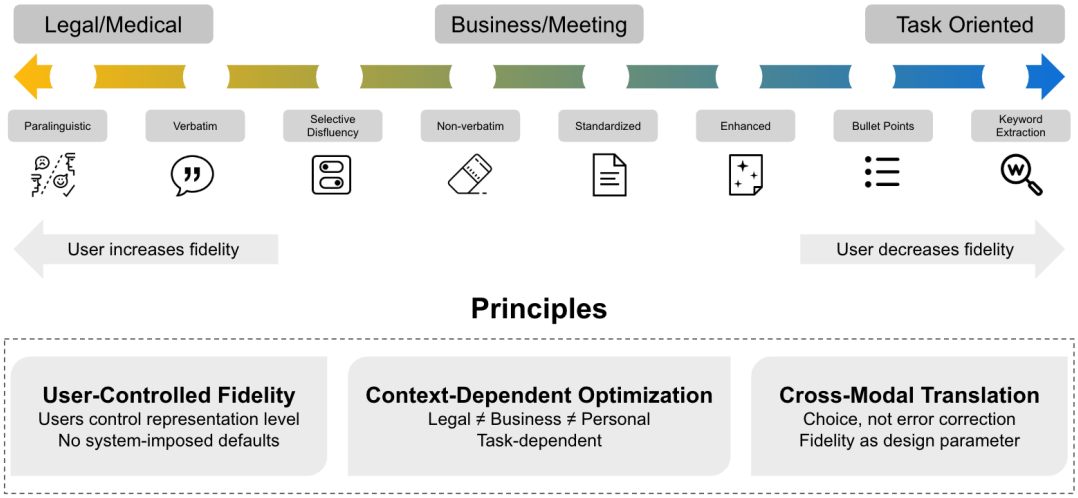
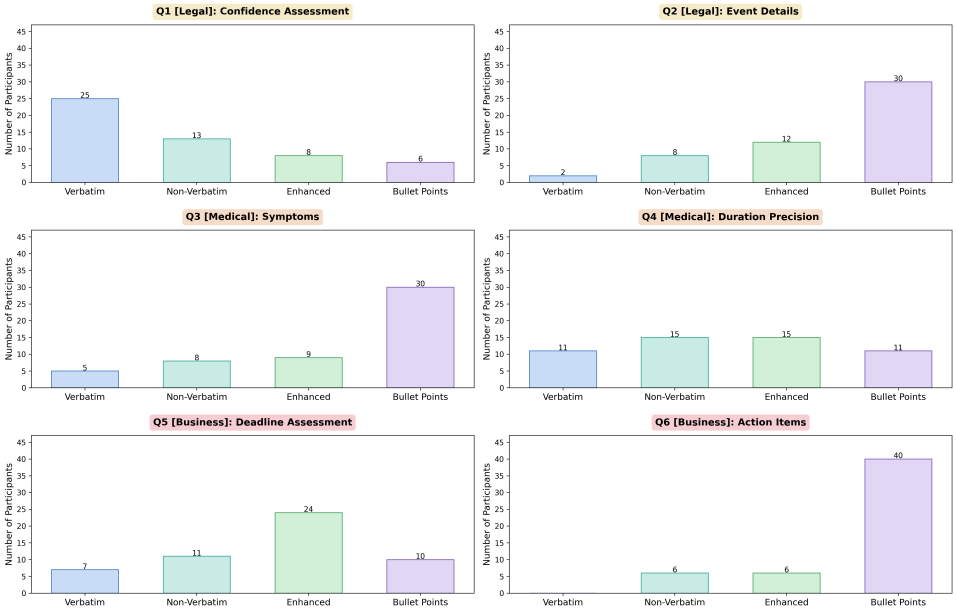


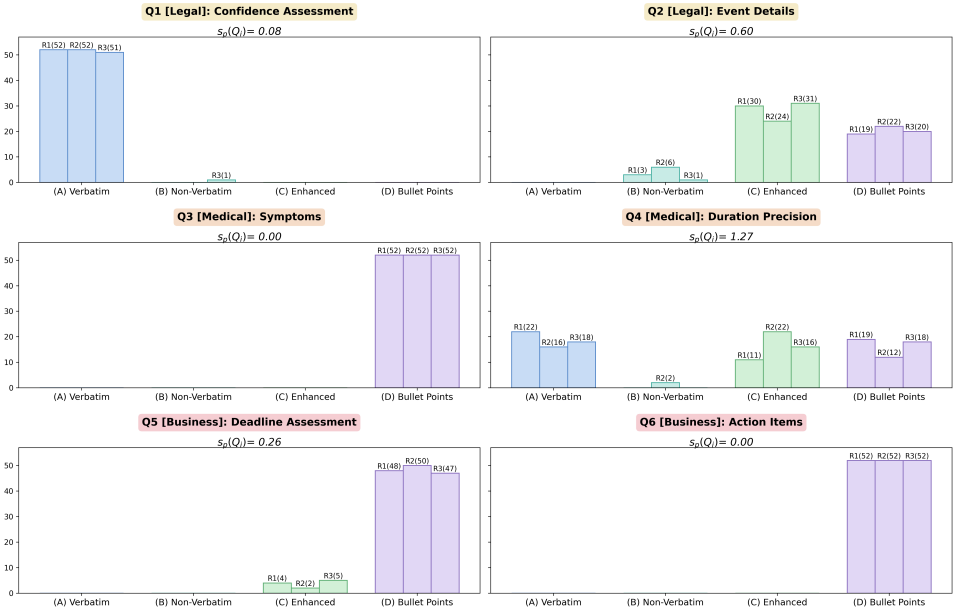
Figure 1: The SpeechSpectrum framework conceptualizes STT conversion as a continuous spectrum of verbatimicity levels rather than a single transcription target.

Human vs. Uniform Preference Distributions						
Q_i [Domain]	χ^2	df	p	V	Most Freq. Representation (p_1)	$\Delta(Q_i)$ [$\Delta(Q_i)CI_{95\%}$]
Q_1 [Legal]	16.77	3	0.0008***	0.33	Verbatim	0.23 [0.02, 0.44]
Q_2 [Legal]	33.54	3	0.0000***	0.46	Bullet Points	0.35 [0.12, 0.56]
Q_3 [Medical]	30.31	3	0.0000***	0.44	Bullet Points	0.40 [0.17, 0.58]
Q_4 [Medical]	1.23	3	0.7456	0.09	Non-Verbatim / Enhanced	-
Q_5 [Business]	13.08	3	0.0045**	0.29	Enhanced	0.25 [0.04, 0.42]
Q_6 [Business]	76.62	3	0.0000***	0.70	Bullet Points	0.65 [0.44, 0.81]
LLM_{R1} vs. Uniform Preference Distributions						
Q_i [Domain]	χ^2	df	p	V	Most Freq. Representation (p_1)	$\Delta(Q_i)$ [$\Delta(Q_i)CI_{95\%}$]
Q_1 [Legal]	460.05	3	0.0000***	0.99	Verbatim	1.00 [1.00, 1.00]
Q_2 [Legal]	127.23	3	0.0000***	0.52	Enhanced	0.21 [0.02, 0.48]
Q_3 [Medical]	468.00	3	0.0000***	1.00	Bullet Points	1.00 [1.00, 1.00]
Q_4 [Medical]	47.64	3	0.0000***	0.32	Verbatim / Enhanced	0.06 [0.00, 0.29]
Q_5 [Business]	386.21	3	0.0000***	0.91	Bullet Points	0.85 [0.69, 0.96]
Q_6 [Business]	468.00	3	0.0000***	1.00	Bullet Points	1.00 [1.00, 1.00]

Table 1: Human preferences are generally diffuse, often exhibiting near-ties among representations (mostly moderate to strong Cramér's V, small $\Delta(Q_i)$), whereas LLM preferences are strongly concentrated with single-choice dominance.



(a) Across six task scenarios, human participants selected different transcript versions depending on task needs, demonstrating that no single fidelity level universally serves all contexts.



(b) When asked to complete the same six scenarios as human participants (Figure 2a), LLMs exhibited more extreme preference patterns, often converging on a single transcript type (e.g., consistently selecting bullet points for business tasks).

Figure 2: Comparison of human and LLM preference distributions across six task scenarios.